MEDINOVA TEACHING HOSPITAL

Class 09 — Renal Medicine

AKI · CKD · Electrolyte Disorders · Dialysis

> *"The kidneys see everything the rest of the body tries to hide."*

SENIOR RESIDENCY SERIES | MODULE 09 of 20

SECTION 01

Acute Kidney Injury (AKI)

AKI is defined by a sudden decrease in GFR, evidenced by an increase in Serum Creatinine (SCr) or a decrease in Urine Output (UO).

KDIGO Staging for AKI

Stage	Creatinine Criteria	Urine Output Criteria
1	1.5 – 1.9x baseline OR ≥ 26.5 µmol/L increase	< 0.5 mL/kg/h for 6–12h
2	2.0 – 2.9x baseline	< 0.5 mL/kg/h for ≥ 12h
3	3.0x baseline OR SCr ≥ 353.6 µmol/L OR RRT start	< 0.3 mL/kg/h for ≥ 24h OR Anuria ≥ 12h

Etiology Differentiation

- **Pre-Renal (60%):** Hypovolaemia, Heart Failure, Sepsis. (Low Na in urine).
- **Intrinsic (35%):** ATN (Acute Tubular Necrosis), Vasculitis, Drugs (NSAIDs, Gentamicin).
- **Post-Renal (5%):** Stones, BPH, Malignancy. (Requires Imaging/Catheter).

> **Prof. Obi's Pearl #1** "The creatinine is a rearview mirror. When it starts to rise, the injury happened yesterday. When it starts to fall, the kidneys actually healed hours ago."

SECTION 02

Hyperkalaemia: Emergency Management

Potassium > 6.5 mmol/L or any K+ with ECG changes is a medical emergency.

ECG Progression

- **K+ 5.5 – 6.5:** Tall, "Tented" T-waves.
- **K+ 6.5 – 7.5:** Loss of P-wave, Prolonged PR interval.
- **K+ > 7.5:** Widened QRS, "Sine Wave" pattern → VF/Asystole.

Treatment Stepladder

Phase	Drug & Dose	Onset / Duration
1. Protect Heart	10ml 10% Calcium Gluconate IV	1-3 mins / 30-60 mins
2. Shift K+	10 units Actrapid in 50ml 50% Dextrose	15-30 mins / 4-6 hours
3. Shift K+ (Adjunct)	Salbutamol 10-20mg Nebulised	30 mins / 2 hours
4. Remove K+	Calcium Resonium OR Dialysis	Delayed / Permanent

> Prof. Obi's Pearl #2 "Calcium gluconate does not lower potassium. It only stops the heart from stopping while you work on the actual problem. Don't stop there."

SECTION 03

Chronic Kidney Disease (CKD)

KDIGO "Heat Map" Staging (G and A)

GFR Stage (G)	Description	Albuminuria (A)	ACR (mg/mmol)
G1: > 90	Normal	**A1:** Normal	< 3
G2: 60-89	Mild decrease	**A2:** Mod. Increase	3 - 30
G3a/b: 30-59	Mod. to Sev. decrease	**A3:** Sev. Increase	> 30
G4: 15-29	Severe decrease	*A higher A-stage increases risk of progression regardless of G-stage.*	
G5: < 15	End-Stage (ESRD)		

Complications & Management

- **Mineral Bone Disease:** High Phosphate, Low Calcium, High PTH. Rx: Phosphate binders.
- **Anaemia:** Deficiency of Erythropoietin (EPO). Rx: Iron studies first, then ESA (Erythropoiesis Stimulating Agents).
- **Renoprotection:** ACEi/ARB (SGLT2i are now first-line for most CKD with albuminuria).

SECTION 04

Hyponatraemia Assessment

Sodium < 135 mmol/L. Always assess **Serum Osmolality** and **Volume Status**.

1. Hypovolaemic: (Diuretics, Vomiting). Rx: Normal Saline.
2. Euvolaemic (SIADH): (Small cell lung CA, SSRIs). Rx: Fluid restriction.
3. Hypervolaemic: (Heart Failure, Cirrhosis). Rx: Diuretics + Fluid restriction.

DANGER: Osmotic Demyelination Syndrome

Correcting Na+ too fast (> 8-10 mmol/L in 24h) can cause "Locked-in Syndrome".

SECTION 05

Renal Replacement Therapy (RRT)

Modality	Pros	Cons
Haemodialysis (HD)	Efficient, supervised.	Access issues, hypotension.
Peritoneal (PD)	Home-based, continuous.	Risk of Peritonitis, glucose load.
CVVH (CRRT)	Ideal for ICU (unstable BP).	Expensive, continuous anticoag.

AEIOU Indications for Emergency Dialysis:
Acidosis (pH < 7.1) | **E**lectrolytes (Refractory K+) | **I**ntoxication (Salicylates/Lithium) | **O**verload (Pulmonary Oedema) | **U**raemia (Pericarditis/Encephalopathy).

SECTION 06

Glomerular Disease Comparison

Feature	Nephrotic Syndrome	Nephritic Syndrome
Proteinuria	Massive (> 3.5g/day)	Moderate (< 3.5g/day)
Haematuria	None / Microscopic	Macroscopic ("Coke-colored")
Edema	Severe (Anasarca)	Mild (Periorbital)
Pathology	Podocyte injury (MCD, FSGS)	Glomerular inflammation (IgA)

SECTION 07

Renal Emergencies

1. Rhabdomyolysis: Trauma/Statins → Muscle breakdown → Myoglobinuria. Look for CK > 10,000. Rx: Massive IV Fluids to keep UO > 200ml/h.

2. Contrast-Induced Nephropathy: Prevention is key. IV Saline 1ml/kg/h for 12h pre and post procedure.

> **Prof. Obi's Pearl #3** "If the patient is edematous but the albumin is normal, look at the heart. If the albumin is low but there's no protein in the urine, look at the liver."

> **Prof. Obi's Pearl #4** "NSAIDs are the 'renal poison' of the 21st century. In a patient with AKI, checking the drug chart is more important than checking the ultrasound."

OFFICIAL SIGN-OFF

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SERIES: Senior Residency Module 09 of 20
DATE: March 2026